

# STAT 340 Applied Methods in Statistics Course: Lecture 1 - Introduction

Aliaksandr Hubin

February 3 2025

## **Applied Methods in Statistics**

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### **Research interests:**

- ▶ Bayesian statistics and machine learning;
- ▶ Latent binary Bayesian neural networks;
- ▶ Bayesian generalized nonlinear models;
- ▶ Advanced mode jumping samplers and other approximation schemes for Bayesian inference.

# Class Schedule

## Lectures

- ▶ **When:** Tuesdays 14:15–16:00
- ▶ **Where:** Room T330
- ▶ **First Lecture:** Feb 3

## Exercises (individual work)

- ▶ **When:** Thursdays 14:15–16:00
- ▶ **Where:** Room BT3A10
- ▶ **First Session:** Feb 5

## Exercises (solutions presentation)

- ▶ **When:** Mondays 10:15–12:00
- ▶ **Where:** Room T132
- ▶ **First Session:** Feb 9

# Practical Information

## Mandatory Activity

- ▶ **Four compulsory quiz assignments**
- ▶ You can be released from a quiz by answering 10 questions correctly during lectures (no penalty for wrong answers).
- ▶ This will work even after failing a quiz.

## Exam

- ▶ **Format:** Multiple choice (100%)
- ▶ **Grading:** A–E or Not Passed
- ▶ **Allowed:** All calculators and specified **written** aids
- ▶ **Date & Time:** **May 18, 09:00–12:00**
- ▶ **Important:** No date changes possible

# Learning Activities (Weekly)

- ▶ **Lectures:** 2 hours
- ▶ **Group Exercises:** 2 hours
- ▶ **Individual Work:** 2 hours

**Total:** 6 hours per week

# Assumed Background: STAT 100

Students should be familiar with:

- ▶ Basic rules of probability (we shall repeat that today)
- ▶ Common probability distributions (we shall repeat that today):
  - ▶ Binomial
  - ▶ Gaussian (Normal)
  - ▶ Student's  $t$
  - ▶ Fisher's  $F$
- ▶ Hypothesis testing:
  - ▶ One-sample  $t$ -tests
  - ▶ Two-sample  $t$ -tests
  - ▶ Paired sample  $t$ -tests
- ▶ One-way Analysis of Variance (ANOVA)
- ▶ Simple linear regression (one predictor variable)
- ▶ Contingency tables

# Software & Data

## Software

- ▶ **R Studio** for R programming language
- ▶ Practical analysis using various statistical methods
- ▶ Theoretical and practical interpretation of results
- ▶ (Other software may be used if methods are available)

## Data

- ▶ Mainly R library: **BIAS.data** (GitHub)  
<https://github.com/thoree/BIAS.data>

# STAT 340 Course Content

## Main Topics

1. **Multiple Linear Regression**
2. **Multivariate Analysis**
  - ▶ Principal Component Analysis (PCA)
3. **Classification Methods**
4. **Clustering Methods**
5. **Generalized Linear Models (GLM)**
  - ▶ Binomial/Bernoulli (logistic) regression
  - ▶ Poisson regression
  - ▶ Mixed models (GLMM): Random effects, repeated measurements



# Required Learning Resources

## Primary Materials

- ▶ **Lecture presentations** (provided)
- ▶ **All exercises** (provided)

## Recommended Textbooks (available online)

**Main Reference: - R Book:** Michael J Crawley, Wiley (2nd ed.)

- Chapters: 1–9 (as needed), 10–13, 17, 19, 25 - Freely available

https:

[//www.cs.upc.edu/~robert/teaching/estadistica/TheRBook.pdf](https://www.cs.upc.edu/~robert/teaching/estadistica/TheRBook.pdf)

**Supplementary (mathematical details): - ISL:** James et al. “An Introduction to Statistical Learning” - Chapters: 1–4, 6.1, 7.1, 10

- Freely available <https://www.statlearning.com>

# Typical Week Structure

| ws | Monday 2.2                                    | Tuesday 3.2                                    | Wednesday 4.2 | Thursday 5.2                                    | Friday 6.2 |
|----|-----------------------------------------------|------------------------------------------------|---------------|-------------------------------------------------|------------|
| 8  |                                               |                                                |               |                                                 |            |
| 9  |                                               |                                                |               |                                                 |            |
| 10 | 11:00                                         |                                                |               |                                                 |            |
| 11 | STAT340<br>T132<br>Hjelpørelser -<br>Exercise |                                                |               |                                                 |            |
| 12 | 13:00                                         |                                                |               |                                                 |            |
| 13 |                                               |                                                |               |                                                 |            |
| 14 |                                               | 15:00                                          |               | 14:00                                           |            |
| 15 |                                               | STAT340<br>T1330<br>Aleksandr Hubin<br>Lecture |               | STAT340<br>B13010<br>Hjelpørelser -<br>Exercise |            |
| 16 |                                               | 16:30                                          |               | 16:00                                           |            |
| 17 |                                               |                                                |               |                                                 |            |

|                         |        | Response   |                      |                |                |                                                       |
|-------------------------|--------|------------|----------------------|----------------|----------------|-------------------------------------------------------|
|                         |        | Continuous |                      | Discrete       |                |                                                       |
|                         |        | Normal     | Non-normal           | Counts         | Bivariate      | Multivariate                                          |
| "Study of associations" |        | "Linear"   | (Transform?)         | Poisson        | Logistic       | Multinomial                                           |
| Predictors              | Fixed  | Contin.    | Linear Regression    | GLM regression | GLM Regression | GLM regression or Discriminant analysis (LDA, QDA ++) |
|                         |        | Discrete   | ANOVA                | GLM ANOVA      | GLM ANOVA      | GLM ANOVA/Contingency tables or Discriminant analysis |
|                         |        | Both       | ANCOVA<br>Dummy regr | GLM ANCOVA     | GLM ANCOVA     | GLM ANCOVA or Discriminant analysis                   |
|                         | Random | Any kind   | LMM                  | GLMM           |                |                                                       |
| New sample              |        | Any kind   | Prediction           |                |                | Classification                                        |

# Reflection Activity

**Turn to your neighbour(s) and discuss:**

- ▶ Do you know what kind of **data** you will be dealing with in your work?
- ▶ Do you have any idea what **association studies** or **methods** are relevant for your research?
- ▶ Are they mentioned in the table in the previous page?

# Now, Let's Begin!

## Lecture 1, Part B: Review of Basic Statistics

We'll start with basics of **probability theory** - the mathematical foundation for **uncertainty** and **statistical inference**.

# Part 0: Foundations of Probability

# Setting Up a Random Experiment

Before studying random variables, we need to define randomness formally.

**A probability setup has three components:**

1. **Sample space  $\Omega$ :** all possible outcomes
  - ▶ *Discrete:* Coin flip:  $\Omega = \{H, T\}$
  - ▶ *Continuous:* Adult height:  $\Omega = [120, 250]$  cm
2. **Events  $\mathcal{F}$ :** set of subsets of outcomes we observe
  - ▶ *Discrete:*  $\{H\} = \text{"heads"}$  or  $\{H\} = \text{"tail"}$
  - ▶ *Continuous:*  $\{h : \text{between 160 and 180}\}$  or  $\{h : \text{between 240 and 250}\}$
3. **Probabilities  $P$ :** assigns to events a number in  $[0, 1]$ 
  - ▶ *Discrete:*  $P(H) = 0.5$
  - ▶ *Continuous:*  $P(160 \leq h \leq 180) = 0.68$  or  $P(240 \leq h \leq 250) = 0.001$

Together:  $(\Omega, \mathcal{F}, P)$  is a **probability space**.

# Andrey Kolmogorov (1903–1987)



Figure 1: Soviet mathematician who revolutionized probability by providing rigorous mathematical foundations

- ▶ Established the **foundations of probability** (1933)
- ▶ Formalized the concept of probability space  $(\Omega, \mathcal{F}, P)$ .

## Kolmogorov's Axioms:

1.  $P(\Omega) = 1$  (certainty has probability 1)
2.  $P(A) \geq 0$  for all outcomes  $A$  (non-negativity)
3. Countable additivity: If  $A_1, A_2, \dots$  are disjoint events, then
$$P(\cup_i A_i) = \sum_i P(A_i)$$

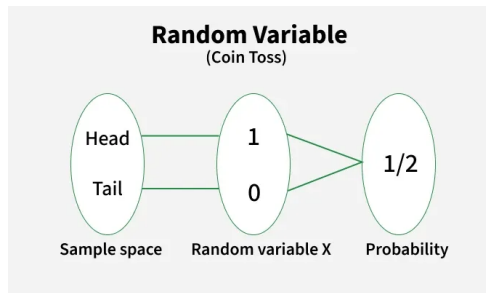


# Part 1: Random Variables

# What is a Random Variable?

A **random variable** is a function that assigns a numeric value to each outcome in the sample space. Maps from experiment sample space to a real space.

**Why?** Numbers let us compute statistics: averages, spreads, and patterns.

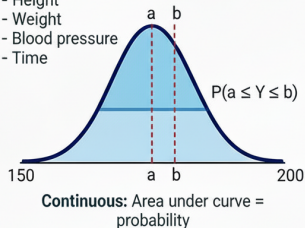


**Example:** Coin flip outcome  $\rightarrow$  Number: - Heads  $\rightarrow Y = 1$  - Tails  $\rightarrow Y = 0$

# Discrete vs. Continuous Random Variables

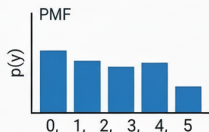
## CONTINUOUS RANDOM VARIABLES

- No distinct categories
- No limit on the value
- Tends to be quantitative
- Height
- Weight
- Blood pressure
- Time



## DISCRETE RANDOM VARIABLES

- Distinct categories (countable)
- No in-between values
- Tends to be qualitative/countable
- Coin flips
- Patient counts at emergency room
- Number of rings in a tree



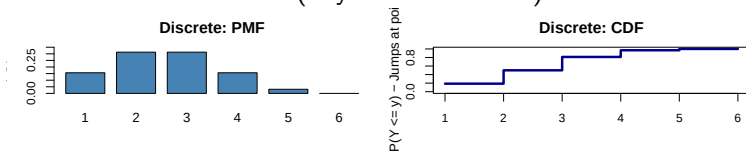
Discrete: Heights of bars = probability at each point

**Continuous:** Infinite uncountably many points  
→ each single point has  $P(Y=y)=0$ . Only  
INTERVALS have non-zero probability.

**Discrete:** Countably many points → each  
point can have  $P(Y=y)>0$ . We sum  
**probabilities** at individual points.

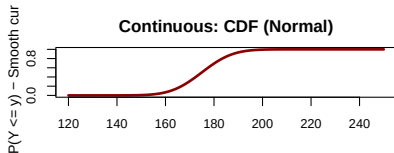
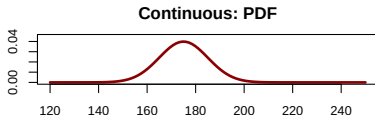
# Discrete Random Variables

- ▶ **Countable** values (often integers)
- ▶ Examples: Coin flips, patient counts at emergency room, number of rings in a tree
- ▶ Use **probability mass function (PMF)**  $p(y)$ :  
 $P(Y = y) = p(y)$  and  $\sum_{\text{all } y} p(y) = 1$
- ▶ **Cumulative distribution function (CDF)**:  
 $F(y) = P(Y \leq y) = \sum_{x \leq y} p(x)$
- ▶ Also exist quantile functions, moment generating functions, characteristic functions (beyond this course).



# Continuous Random Variables

- ▶ Values in an **interval** (uncountably many)
- ▶ Examples: Height, weight, blood pressure, time
- ▶ Use **probability density function (PDF)**  $f(y) \geq 0$
- ▶ Probabilities:  $P(a \leq Y \leq b) = \int_a^b f(y) dy$  and  $\int_{-\infty}^{\infty} f(y) dy = 1$
- ▶ **Cumulative distribution function (CDF):**  
 $F(y) = P(Y \leq y) = \int_{-\infty}^y f(x) dx$



## CRITICAL DIFFERENCE: Single Point Probability

**CHECKPOINT:** Why does  $P(Y = 172.5) = 0$  for continuous height?

| Property                | Discrete                          | Continuous                     |
|-------------------------|-----------------------------------|--------------------------------|
| $P(Y = y)$              | <b>Can be <math>&gt; 0</math></b> | <b>Always = 0</b>              |
| Example: Fair die       | $P(Y = 3) = 1/6$                  | —                              |
| Example: Height         | —                                 | $P(\text{height} = 172.5) = 0$ |
| How to find probability | Use $P(Y = y)$                    | Use $P(a \leq Y \leq b)$       |

**Intuition:** Continuous distributions have *uncountably many* points (infinity!), so each single point contributes zero probability. Only **intervals with area** have non-zero probability.

## Expectation: Discrete Case

For a **discrete**  $Y$  with probabilities  $p(y)$ :

$$E(Y) = \mu = \sum_{\text{all } y} y \cdot p(y)$$

**Fair die example** ( $\{1, 2, 3, 4, 5, 6\}$ , each with  $p(y) = 1/6$ ):

$$E(Y) = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} = 3.5$$

**Interpretation:** Long-run average over many rolls.

## CHECKPOINT: Understanding $E(Y)$ for the Fair Die

**Question:** Why is  $E(Y) = 3.5$ , not 3 or 4? What does 3.5 mean?

- ▶ Is 3.5 a possible outcome when rolling a die? **NO**
- ▶ Does it mean we expect to see 3.5 on the next roll? **NO**
- ▶ Does it mean over 1,000 rolls, the average is *approximately* 3.5? **YES**

**Key insight:**  $E(Y)$  is the **balance point** of the distribution. Imagine the die outcomes as weights on a seesaw—3.5 is where it balances!

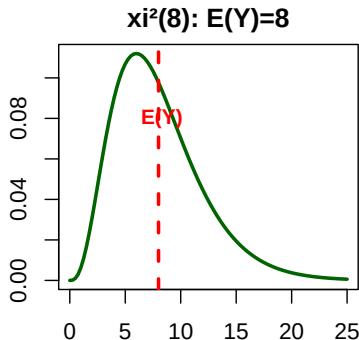
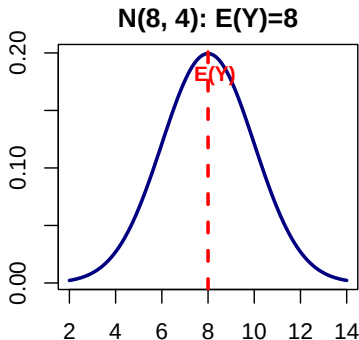


## Expectation: Continuous Case

For a **continuous**  $Y$  with density  $f(y)$ :

$$E(Y) = \mu = \int_{-\infty}^{\infty} y f(y) dy$$

**Interpretation:** Center of mass (balance point) of the distribution.



## Variance: Discrete Case

For a **discrete**  $Y$ :

$$\text{Var}(Y) = \sigma^2 = E((y - \mu)^2) = \sum_{\text{all } y} (y - \mu)^2 \cdot p(y)$$

**Fair die with**  $\mu = 3.5$ :

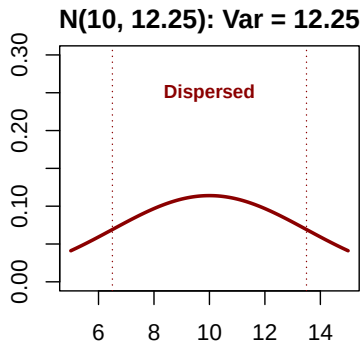
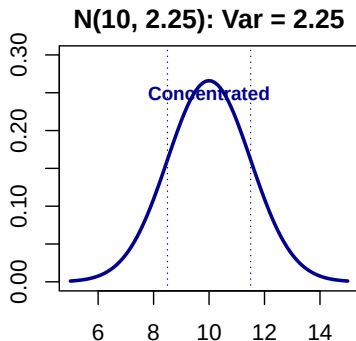
$$\text{Var}(Y) = (1 - 3.5)^2 \cdot \frac{1}{6} + \dots + (6 - 3.5)^2 \cdot \frac{1}{6} \approx 2.92$$

## Variance: Continuous Case

For a **continuous**  $Y$ :

$$\text{Var}(Y) = \sigma^2 = \int_{-\infty}^{\infty} (y - \mu)^2 f(y) dy$$

**Standard deviation:**  $\sigma = \sqrt{\text{Var}(Y)}$  (same units as  $Y$ )



**Larger variance = more spread!**

# Sample Mean and Standard Deviation

Given  $n$  observations  $y_1, \dots, y_n$ :

**Sample mean** (estimates  $\mu$ ):

$$\bar{y} = \frac{1}{n} \sum_{t=1}^n y_t$$

**Sample standard deviation** (estimates  $\sigma$ ):

$$s = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (y_t - \bar{y})^2}$$

We divide by  $(n - 1)$  for unbiased variance estimation (**Bessel's correction**).

## Part 5: Rules for Transformations

## Linear Transformations: $E(cY)$ and $Var(cY)$

For constant  $c$  and random variable  $Y$ :

$$E(c) = c,$$

$$E(cY) = c \cdot E(Y)$$

$$Var(c) = 0,$$

$$Var(cY) = c^2 \cdot Var(Y)$$

**Key insight:** Variance scales with  $c^2$ . Doubling a variable multiplies variance by 4!

CHECKPOINT: can someone say how to prove  $\text{Var}(cY) = c^2\text{Var}(Y)$ , not  $c\text{Var}(Y)$ ?

**Question:** If we triple the values (multiply by 3), variance multiplies by 9. Why the square?

# Sums of Independent Variables

For independent  $Y_1, \dots, Y_n$  with expectations  $\mu_i$  and variances  $\sigma_i^2$  and some constants  $c_i$ :

**Expectation of sum:**

$$E \left( \sum_{i=1}^n c_i Y_i \right) = \sum_{i=1}^n c_i \mu_i$$

**Variance of sum (independent only):**

$$\text{Var} \left( \sum_{i=1}^n c_i Y_i \right) = \sum_{i=1}^n c_i^2 \sigma_i^2$$



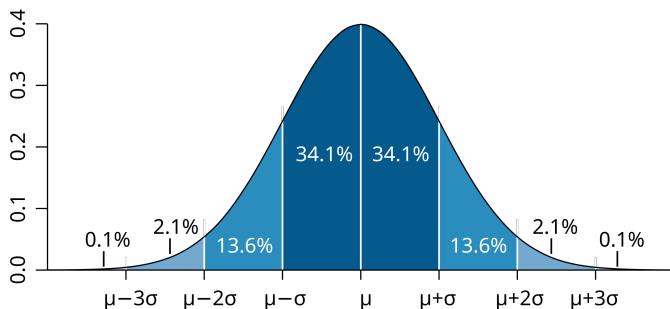
## Part 6: Key Distributions

# The Normal Distribution

**Notation:**  $Y \sim N(\mu, \sigma^2)$

**PDF:**  $f(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right\}$

**Empirical Rule & Visual:**



## CHECKPOINT: SAT Scores and the 68-95 Rule

**Application:** SAT scores follow  $N(\mu = 500, \sigma^2 = 100^2)$ .

- ▶ What % score between **400-600**? (within 1 standard deviation)
- ▶ What % score between **600-700**? (1 to 2 standard deviations away from mean)

## CHECKPOINT: SAT Scores and the 68-95 Rule

**Application:** SAT scores follow  $N(\mu = 500, \sigma = 100)$ .

- ▶ What % score between **400-600**? (within 1 standard deviation)
  - ▶ **Answer: 68.2%**
- ▶ What % score between **300-700**? (2 standard deviations away from mean)
  - ▶ **Answer: 95.4%**

**Why this matters:** Standardized tests are designed using normal distributions. Knowing the 68-95 rule helps interpret scores!

## Carl Friedrich Gauss (1777–1855)

The “**Prince of Mathematicians**” and namesake of the normal distribution.



Figure 2: German mathematician with groundbreaking contributions to statistics (**error theory** back then), calculus, algebra and number theory, and astronomy

# Linear Combinations of Normal Variables

If  $Y_1 \sim N(\mu_1, \sigma_1^2)$  and  $Y_2 \sim N(\mu_2, \sigma_2^2)$  are *independent*:

$$aY_1 + bY_2 \sim N(a\mu_1 + b\mu_2, a^2\sigma_1^2 + b^2\sigma_2^2)$$

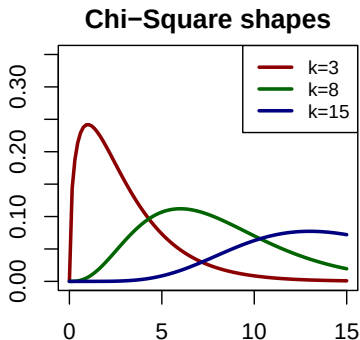
This is **fundamental** for inference about sample means and regression.

# The Chi-Square Distribution

**Notation:**  $Y \sim \chi_k^2$  where  $k$  = degrees of freedom

**Key derivation:**  $Z^2 \sim \chi_1^2$  if  $Z \sim N(0, 1)$ ; additive property:  
 $Y = \sum_{i=1}^k Z_i^2 \sim \chi_k^2$

**Properties:**  $E(Y) = k$ ,  $Var(Y) = 2k$

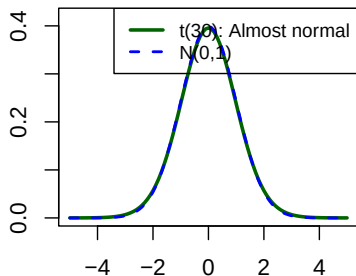
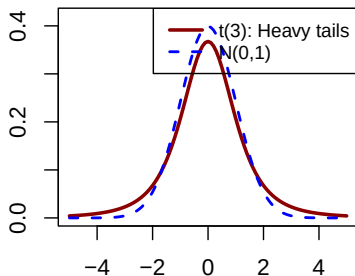


# The Student's $t$ Distribution

**Definition:** For independent  $Z \sim N(0,1)$  and  $Y \sim \chi_k^2$ :

$$T = \frac{Z}{\sqrt{Y/k}} \sim t_k$$

**t-distribution with small df    t-distribution converges to normal**



**Properties:** Symmetric, heavier tails than normal, converges to  $N(0,1)$  as  $k \rightarrow \infty$



## William Sealy Gosset (1876–1937): “Student”



Figure 3: English scientist working as a chemist for Guinness Brewery. Developed the  $t$  distribution to handle small samples when variance is unknown (1908)

- ▶ Faced a practical problem: small sample inference for quality control
- ▶ Published under the pseudonym “**Student**” because Guinness forbade employee publications

# The $t$ Distribution: Practical Use

**Test statistic for population mean:**

If  $y_1, \dots, y_n$  i.i.d.  $N(\mu, \sigma^2)$  and  $S^2 = \frac{1}{n-1} \sum_{t=1}^n (y_t - \bar{y})^2$ :

$$T = \frac{\bar{Y} - \mu}{\sqrt{S^2/n}} \sim t_{n-1}$$

This is the **one-sample**  $t$  test statistic.

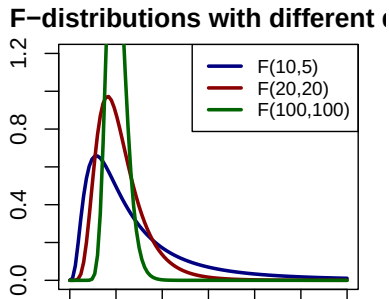
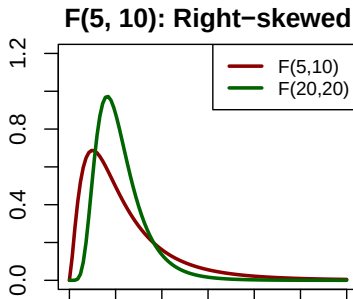
# The $F$ Distribution

**Notation:**  $Y \sim F_{u,v}$  where  $u, v = \text{degrees of freedom}$

**Definition:** For independent  $\chi_u^2$  and  $\chi_v^2$ :

$$F = \frac{\chi_u^2/u}{\chi_v^2/v} \sim F_{u,v}$$

$$E(Y) = \frac{v}{v-2}, v > 2 \quad \text{Var}(Y) = \frac{2v^2(u+v-2)}{u(v-2)^2(v-4)} v > 4$$



## Ronald Aylmer Fisher (1890–1962)



Figure 4: British statistician and geneticist with revolutionary impact on statistics and evolutionary biology. The founder of modern statistics and namesake of the F-distribution.

- ▶ Invented the **F distribution**, **ANOVA** (analysis of variance), and **maximum likelihood estimation**
- ▶ Pioneered **experimental design** principles (randomization, blocking, replication)
- ▶ Won the prestigious **Weldon Medal** for his contributions to statistical science

**Legacy:** Fisher's work transformed statistics from a descriptive tool to a framework for scientific inference.

Many more distributions of course

